

Q2: How many solutions?

$$ax^2+bx+c\equiv 0\ (a\neq 0)$$









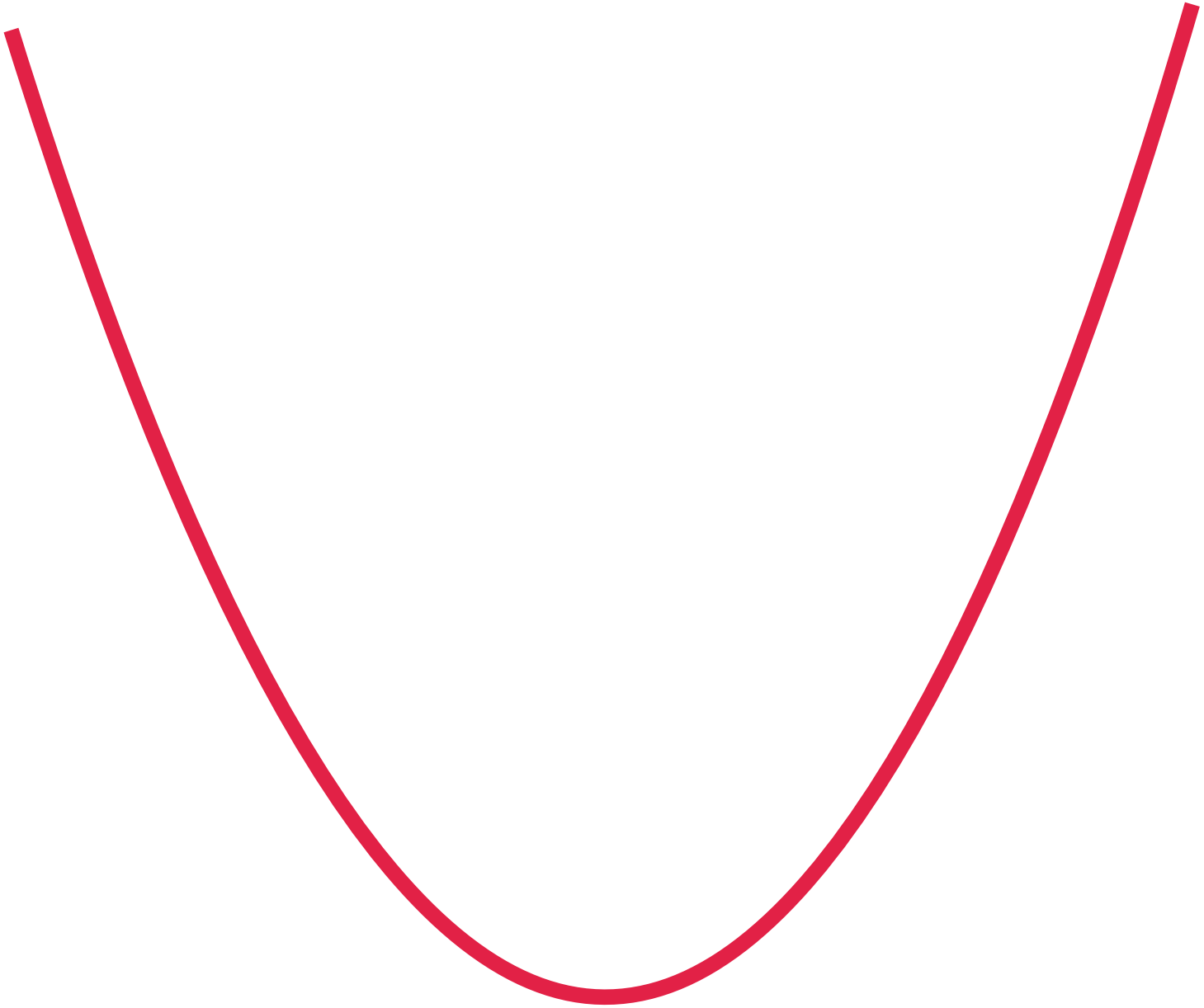


Number of *real* solutions











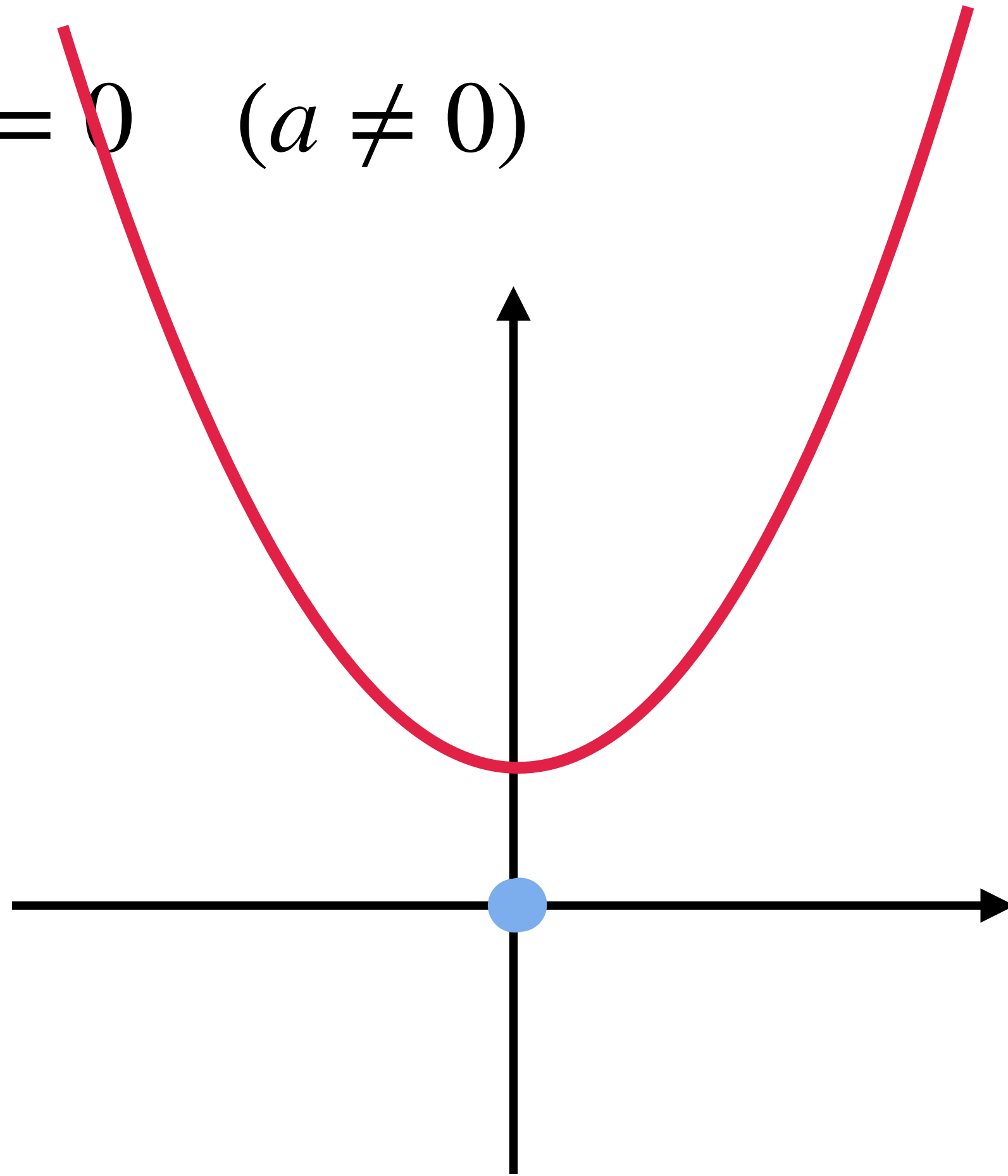


$$\Delta = b^2 - 4ac$$



Q2: How many solutions?

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$



$$\Delta = b^2 - 4ac$$

Number of *real* solutions

$$\Delta > 0$$

2

$$\Delta = 0$$

1²

$$\Delta < 0$$

0

Q2: How many solutions?

$$\Delta = b^2 - 4ac$$

$$x_1, x_2 = \frac{-b \pm \sqrt{\Delta}}{2a}$$